FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2021

PAPER –1

TEST DATE: 01-08-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases.

Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

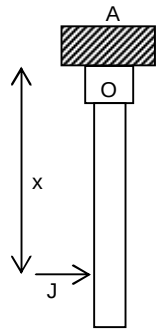
This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A ball is projected from a point in one of the two smooth parallel vertical walls against the other in a plane perpendicular to both. After being reflected at each wall impinge again on the second at a point in the same horizontal plane as it started from. If "e" is the coefficient of restitution, "a" is the distance between the walls and "R" is the free range on horizontal surface, then which of the following is correct?
   (A) $Re^3 = a(1 + e + e^2)$
   (B) $Re = a(1 + e^2 + e^3)$
   (C) $Re^2 = a(1 + e + e^2)$
   (D) $R = a(1 + e + e^2)$

2. A hemisphere rests in equilibrium on a rough ground and against an equally rough wall, if the equilibrium is limiting, then the inclination θ of the base to the horizontal is
   (A) $\theta = \sin^{-1}\left[\frac{3}{8}\left\{\frac{\mu+\mu^2}{1+\mu^2}\right\}\right]$
   (B) $\theta = \sin^{-1}\left[\frac{8}{3}\left\{\frac{\mu+\mu^2}{1+\mu^2}\right\}\right]$
   (C) $\theta = \cos^{-1}\left[\frac{8}{3}\left\{\frac{\mu+\mu^2}{1+\mu^2}\right\}\right]$
   (D) $\theta = \cos^{-1}\left[\frac{3}{8}\left\{\frac{\mu+\mu^2}{1+\mu^2}\right\}\right]$

3. A cubical container of side 'a' and wall thickness x (x << a) is suspended in air and filled n moles of diatomic gas (adiabatic exponent γ) in a room where room temperature is T_0. If at time t = 0 gas temperature is T_1 ($T_1 > T_0$), the temperature of gas 'T' at time 't' is
   (A) $T = T_0 + T_1 e^{-\frac{6ka^2(\gamma-1)}{nRx}t}$
   (B) $T = T_0 - (T_1 - T_0)e^{-\frac{6ka^2(\gamma-1)}{nRx}t}$
   (C) $T = T_0 + (T_1 + T_0)e^{-\frac{6ka^2(\gamma+1)}{nRx}t}$
   (D) $T = T_0 + (T_1 - T_0)e^{\frac{6ka^2(\gamma-1)}{nRx}t}$

4. A uniform rod of length 'l' is pivoted at point 'A'. It is struck by a horizontal force which delivers an impulse 'J' at a distance 'x' from point 'A' as shown in figure, impulse delivered by pivot is zero if 'x' is equal to



(A) $\frac{l}{2}$

(B) $\frac{l}{3}$

(C) $\frac{2l}{3}$

(D) $\frac{3l}{4}$

5. A planet revolves around the sun in an elliptical orbit of eccentricity *e*. If *T* is the time period of the planet, then the time spent by the planet between the ends of the minor axis and major axis close to the sun is

(A) $T\left(\frac{1}{4}-\frac{e}{2\pi}\right)$

(B) $\frac{T.e}{2\pi}$

(C) $T\left(\frac{2e}{\pi}-1\right)$

(D) $\frac{T\pi}{2e}$

6. Force acting on a particle moving in a straight line varies with the velocity of the particle as $F=\frac{k}{v}$, where k is a constant. The work done by this force in time t is:

(A) $\frac{k}{v^2}t$

(B) 2kt

(C) kt

(D) $\frac{2kt}{v^2}$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. A particle is projected from 'O' at an elevation 'α'. Then which of the fallowing statement/statements is/are correct of the particle.

(A) There may be two positions of 'A' such that OA is perpendicular to the instantaneous velocity.

(B) For real positions to exist, α can not be less than $\cos^{-1}\left(\frac{1}{3}\right)$

(C) For real positions to exist, α can not be less than $\cos^{-1}\left(\frac{2}{3}\right)$

(D) If in the two positions OA makes angles θ_1 and θ_2 with the horizontal then $\theta_1+\theta_2=\alpha$.

8. A long cable is constructed of a solid conductor of radius 'r' in a co-axial hollow cylindrical conductor of inner and outer radii R_1 and R_2 respectively. I_0 current flows in both the conductors in opposite directions. The magnetic field B at a distance x from the axis of the cable is

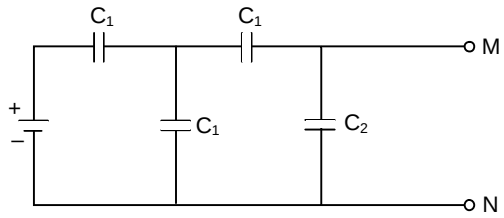
(A) $B = \frac{\mu_0 I_0}{2\pi x'}$ when $r < x < R_1$

(B) $B = \frac{\mu_0 I_0}{2\pi r^{2'}}$ when $x < r$

(C) $B = \frac{\mu_0 I_0 \left(R_2^2 - x^2\right)}{2\pi x\left(R_2^2 - R_1^2\right)'}$ when $R_1 < x < R_2$

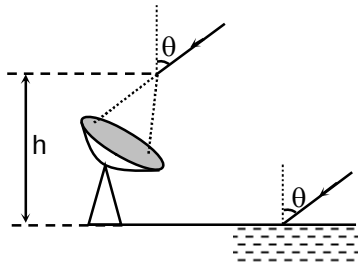
(D) $B = 0$, when $x > R_2$

9. If the capacitance ratio $\frac{C_2}{C_1} = \eta$, then the potential difference V between points M and N in the given figure is



(A) If E=110V and η = 2 then V= 10 volt
(B) If E=285V and η = 3 then V= 15 volt
(C) If E=110V and η = 1/2 then V= 10 volt
(D) If E=380V and η = 3 then V= 20 volt

10. Radio waves of wavelength λ at an angle θ to vertical are received by a radar after reflecting from a nearby water surface and directly. If the radar records a maximum intensity, the height of antenna *h* from water surface can be



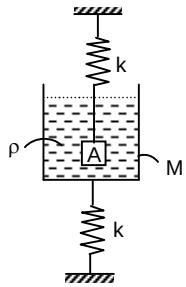
(A) $\frac{5\lambda}{4\cos\theta}$

(B) $\frac{\lambda}{2\sin\theta}$

(C) $\frac{\lambda}{4\sin\theta}$

(D) $\frac{\lambda}{4\cos\theta}$

11. The system shown in the adjacent figure is in equilibrium. The mass of the container with liquid is M, density of liquid in the container is ρ and volume of the block is V. If the container is now displaced downwards through a distance x_0 and released such that the block remains well inside the liquid, then during subsequent motion



(A) time period of SHM of the container will be $2\pi\sqrt{\frac{M}{k}}$

(B) time period of SHM of the container will be $2\pi\sqrt{\frac{(M+\rho v)}{k}}$

(C) amplitude of SHM of the container will be greater than x_0

(D) container will not perform SHM

12. Nucleus *A* decays to *B* with decay constant λ_1 and *B* decays to *C* with decay constant λ_2. Initially at $t = 0$, number of nuclei of *A* and *B* are $2N_0$ and N_0 respectively. At $t = t_0$, number of nuclei of *B* stop changing. If at this instant number of nuclei of *B* are $\frac{3N_0}{2}$.

(A) the value of t_0 is $\frac{1}{\lambda_1}\ln\frac{4}{3}\frac{\lambda_1}{\lambda_2}$

(B) the value of t_0 is $\frac{1}{\lambda_2}\ln\frac{4}{3}\frac{\lambda_1}{\lambda_2}$

(C) the value of N_A at t_0 is $\frac{3N_0}{2}\frac{\lambda_2}{\lambda_1}$

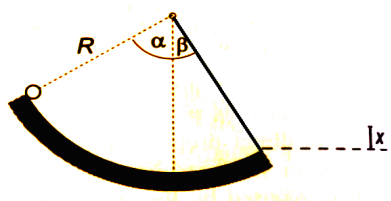
(D) the value of N_A at t_0 is $\frac{2N_0}{3}\frac{\lambda_2}{\lambda_1}$

SECTION – C
(Numerical Answer Type)

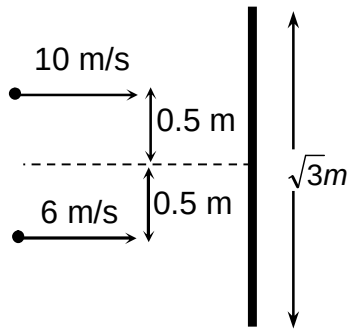
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A solid sphere of radius R = 0.2 m and of negligible mass is swimming in the water tank of base surface area A = 0.5 m^2. The depth of the water in the tank is h = 1 m. Determine the work (in joules) needed to push the sphere down the bottom of tank. (Assume that no water flows out of the tank. The density of water is ρ = 1000 kg/m^3 and g = 9.81 m/s^2).

14. A small wheel, initially at rest, rolls down a ramp in the shape of a quarter circle without slipping. The radius of the circle is R = 1 m and α = 60°, β = 30° . Find the height 'x' (in cm) reached by the wheel after leaving the track.



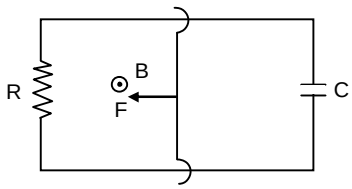
15. A thin uniform bar lies on a frictionless surface and is free to move in any way on the surface. Its mass is 0.16 kg and length $\sqrt{3}$ metres. Two particles each of mass 0.08 kg are moving on the same surface and towards the bar in a direction perpendicular to the bar, one with a velocity 10 m/s and the other with 6 m/s as shown in Figure. The first particle strikes the bar at *A* and the other at the point B. The points A and B are at distances 0.5 m from centre of the bar. The particles strike the bar at the same instant of time and stick to the bar on collision. Calculate the loss of kinetic energy of the system on the above collision process.



16. The main scale of vernier calipers reads in mm and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. It was also observed when two jaws are brought in contact the zero of the vernier scale coincided with the zero of the main scale. The length of a cube is measured with this vernier calipers. The main scale reads 10 mm and the first division of the vernier scale coincides with the main scale. If the mass of the cube is 2.7369 g then, find the density of the cube in (gm/cm^3) upto appropriate significant figures.

17. In a thermally isolated system, two boxes filled with an ideal gas are connected by a valve. When the valve is in closed position, states of the box 1 and 2, respectively, are (1 atm, V, T) and (0.5 atm, 4V, T). When the valve is opened, what is approximately the final pressure (in atm) of the system?

18. Two long parallel rails separated by a distance 'ℓ' are connected by a movable connector of mass 'm'. Terminals of the rails are connected across a resistor 'R' and a capacitor C as shown in figure. A uniform magnetic field B perpendicular to the plane of rails is switched on and the connector is dragged by a constant force F. The terminal velocity (in m/s) of the connector is (Neglect the friction between rails & connector and take $B\ell = 2\sqrt{FR}$).



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

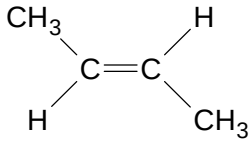
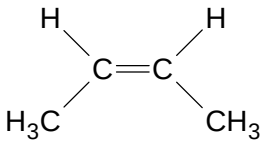
19. The pyknometric density of sodium chloride crystal is 2.165×10^3 kg m^{-3}, while its x-ray density is 2.178×10^3 kg m^{-3}. The fraction of the unoccupied sites in sodium chloride crystal is
(A) 5.96
(B) 5.96×10^{-2}
(C) 5.96×10^{-1}
(D) 5.96×10^{-3}

20. When limestone is heated, quick lime is formed according to reaction
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

The experiment was carried out in temperature range of 800–900°C. Equilibrium constant K_p follows the reaction
$\log K_p = 10 - 8500 / T$
where T is temperature in Kelvin. At what temperature, the above equilibrium can be established if the equilibrium pressure is 1 atm.
(A) 700 K
(B) 750 K
(C) 800 K
(D) 850 K

21. What weight of $H_2C_2O_4.2H_2O$ (molecular weight = 126) should be dissolved to prepare 250 ml of deci-normal solution to be used as reducing agent?
(A) 0.63 gm
(B) 0.1575 gm
(C) 0.126 gm
(D) 0.875 gm

22. $CH_3 - C \equiv C - CH_3 \xrightarrow[Pd-BaSO_4,\ Quinoline]{H_2}$ Product.

The product of above reaction is:
(A) 
(B) $CH_3CH_2C \equiv CH$
(C) 
(D) $CH_2 = CH - CH = CH_2$

23. Which of the following reaction does not produce NH_3 gas?

(A) $Mg_3N_2 + H_2O \longrightarrow$

(B) $NH_4Cl + NaOH \xrightarrow{Heat}$

(C) $(NH_4)_2 Cr_2O_7 \xrightarrow{Heat}$

(D) $NH_4Cl \xrightarrow{Heat}$

24. If ΔH is the enthalpy change and ΔE the change in internal energy accompanying a gaseous reaction then

(A) ΔH is always less than ΔE

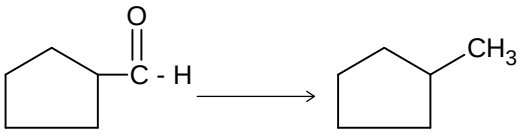
(B) ΔH is always greater than ΔE

(C) ΔH is less than ΔE if the number of moles of gaseous products is greater than the number of moles of gaseous reactants.

(D) ΔH is less than ΔE if the number of moles of gaseous products is less than the number of moles of gaseous reactants.

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. 

The above transformation can be carried out by following methods?

(A) Zn(Hg)/Conc HCl

(B) NH_2NH_2, OH^-

(C) $\begin{matrix} CH_2SH \\ | \\ CH_2SH \end{matrix}$, Dry HCl and Raney Ni, Δ

(D) $LiAlH_4$

26. Silicate having one monovalent corner oxygen atom in each tetrahedron unit is

(A) Sheet silicate

(B) Cyclic silicate

(C) Single chain silicate

(D) Double chain silicate

27. Which of the following species exhibits the paramagnetic behaviour?

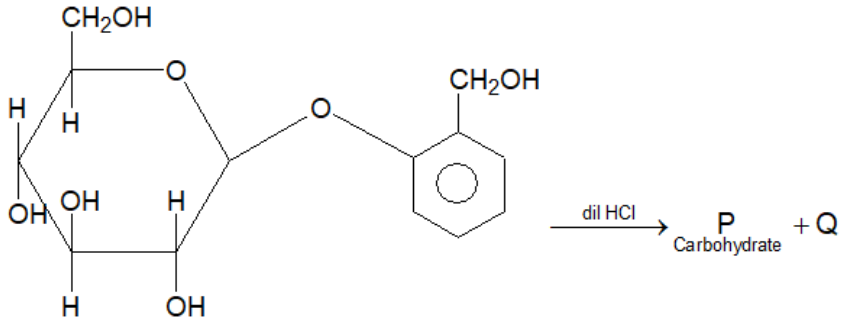
(A) NO

(B) O_2^{2-}

(C) O_2^{2+}

(D) O_2

28. Salicin (structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain.
Observe the following reaction of salicin.



The correct statement(s) is (are) :
(A) P is D-Glucose
(B) Q is 2-hydroxybenzyl alcohol
(C) Q can be converted to a modern analgesic (pain-killer), aspirin
(D) The above reaction occurs through a carbocation

29. 80 gm salt of weak base & strong acid XY is dissolved in water and formed 2 litre of aqueous solution. The pH of the resultant solution was found to be 5 at 298K. If XY forms CsCl type crystal having r_{x^+} =1.6Å & r_{y^-} = 1.864Å

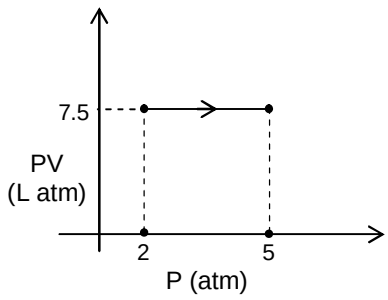
[Given : K_b (XOH) = 4×10^{-5} : $N_A = 6 \times 10^{23}$]
(A) Molar mass of salt is 100 gm/mol
(B) % degree of hydrolysis of salt is 0.25
(C) Edge length of XY is 4Å
(D) Density of Solid salt XY is 2 in gm/cc

30. The rate law for the reaction
RCl+NaOH (aq) $\longrightarrow$ ROH + NaCl, is given by, Rate=K_1[RCl]. The rate of reaction will not be
(A) doubled on doubling the concentration of sodium hydroxide
(B) halved on reducing the concentration of alkyl halide by one half
(C) unaffected by changing the concentration of alkyl halide
(D) unaffected by increasing the temperature of the reaction

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. Work done (in L.atm) for 1 mole of ideal gas which undergoes a reversible process as shown in the graph is



32. For a given reaction energy of activation for forward reaction (E_a) is 80 kJ mol^{-1}. ΔH = -40 kJ mol^{-1} for the reaction A catalyst lowers this E_a to 20 kJ mol^{-1}. The ratio of energy of activation for reverse reaction before and after addition of catalyst is

33. 400 mL of a cylinder contains an ideal gas at 100 K and 2 atm pressure. How much pressure will the same gas exert if it is completely transferred to a 1.6 litre cylinder at same temperature?

34. In diamond, carbon atom occupy fcc lattice points as well as alternate tetrahedral voids. If edge length of the unit cell is 356 pm, then radius of carbon atom is.

35. $NaH_2PO_4 \xrightarrow{>240^\circ C} (NaPO_3)_3 \xrightarrow[\text{(Liquid melt)}]{625^\circ C} NaPO_3 \xrightarrow[\text{Cooling}]{\text{Rapid}} (A)$

If x is the number of oxygen, y is the number of phosphorus and z is the number of sodium atoms in A, then find out $\frac{x}{y+z}$

36. Li, Na, K, Rb, Cs, Ca, Sr, Ba are the s-block elements. If x number of them can produce ammoniated cation and ammoniated electron with liquid ammonia. Then find x/5.

Mathematics

PART – III

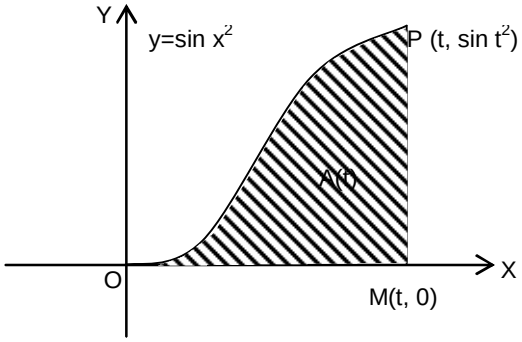
SECTION – A
(One Options Correct Type)

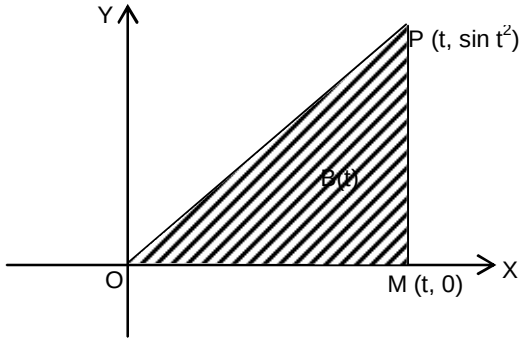
This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If f (x) = $x^4 \tan(x^3) - x \ln(1 + x^2)$, then the value of $\frac{d^4(f(x))}{dx^4}$ at x = 0 is

(A) 0
(B) 6
(C) 12
(D) 24

38. The figure shows two regions in the first quadrant.





A(t) is the area under the curve , $y = \sin x^2$ from 0 to t and B (t) is the area of the triangle with vertices O, P and M (t, 0).Then $\lim_{t\to 0}\frac{A(t)}{B(t)}$ equals

(A) $\frac{2}{3}$
(B) $\frac{3}{2}$
(C) $\frac{1}{2}$
(D) 1

39. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a}\times\left(\vec{a}\times\vec{c}\right)+\vec{b}=0$, then incorrect statement is

(A) $\vec{b}$ is in the plane of $\vec{a}$ and $\vec{c}$
(B) $\vec{b}$ is parallel to $\vec{a}\times\vec{c}$
(C) angle between $\vec{a}$ and $\vec{c}$ is 30°
(D) angle between $\vec{a}$ and $\vec{c}$ is 150°

40. Which of the following function is not twice differentiable at x = 0?

(A) $g(x) = [x^2]\tan^{-1}x - \{x^2\}\cot^{-1}x - [x^2]\frac{\pi}{2}$

(B) $h(x) = |\sin^2 x|$

(C) $h(x) = \begin{cases} x^4\cos\left(\frac{1}{x}\right); & x \neq 0 \\ 0 \quad\quad\quad\quad ; & x = 0 \end{cases}$

(D) None of these

41. Let a, b, c are distinct real numbers, not equal to one. If $ax + y + z = 0$, $x + by + z = 0$ and $x + y + cz = 0$ has a non – trivial solution, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to:

(A) –1
(B) 1
(C) zero
(D) none of these

42. The volume enclosed by the planes $|3x - 4| + |2y - 3| + |z + 4| = 3$ is

(A) 4
(B) 6
(C) 8
(D) None of these

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. The volume of the parallelepiped whose coterminous edges are represented by the vectors $2\vec{b}\times\vec{c}$, $3\vec{c}\times\vec{a}$ and $4\vec{a}\times\vec{b}$, where $\vec{a} = (1+\sin\theta)\hat{i} + \cos\theta\hat{j} + \sin 2\theta\hat{k}$,

$\hat{b} = \sin\left(\theta + \frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta + \frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta + \frac{4\pi}{3}\right)\hat{k}$,

$\vec{c} = \sin\left(\theta - \frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta - \frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta - \frac{4\pi}{3}\right)\hat{k}$ is 18 cubic units, then the value of θ, in the interval $\left(0, \frac{\pi}{2}\right)$, is are

(A) $\frac{\pi}{9}$

(B) $\frac{2\pi}{9}$

(C) $\frac{\pi}{3}$

(D) $\frac{4\pi}{3}$

44. A player tosses a coin. He sets one point for head and two points for tail. He plays till he gets sum of points equal to n. If p_n be the probability that his score becomes n, then

(A) $p_3 = \frac{1}{2}$

(B) $p_n = \frac{1}{2}P_{n-1} + \frac{1}{4}P_{n-2}$

(C) $p_4 = \frac{11}{16}$

(D) $p_n = \frac{1}{2}(p_{n-1} + p_{n-2})$

45. If A and B are two events such that $P(A) = 3/4$ and $P(B) = 5/8$, then

(A) $P(A \cup B) \geq 3/4$

(B) $P(A' \cap B) \leq 1/4$

(C) $3/8 \leq P(A \cap B) \leq 5/8$

(D) None of these

46. If $x \in R$, then the roots of the equation $x^4 + 4x^3 - 8x^2 + k = 0$ with respect to the values of k, $(k \in [0, 2014])$, are :

(A) for $k \in [0,3]$: 4 real and distinct roots

(B) for $k \in (3,128)$: two real and two imaginary roots

(C) for $k \in (128,583)$: no real root

(D) for $k \in (0,3)$: four real and distinct roots

47. If $f(x) = 2x - \sin x$ and $g(x) = \sqrt[3]{x}$, then

(A) range of gof is R

(B) gof is one – one

(C) both f and g are one – one

(D) both f and g are onto

48. $\lim_{x\to\infty} \sqrt{x}\left(\sqrt{x+1} - \sqrt{x}\right)$ equals

(A) $\lim_{x\to 0} \frac{\ln(1+x) - x}{x^2}$

(B) $\lim_{x\to 0} \frac{1 - \cos x}{x^2}$

(C) $\lim_{x\to 0} \frac{\sqrt{1+x} - 1}{x}$

(D) $\lim_{x\to 0} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x^2 + 2x}}}$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If $|z - 3| = \text{Re } z$, $|w - 3| = \text{Re } w$ and $\arg(z - w) = \frac{\pi}{4}$, then find the value of $\text{Im}(z + w)$.

50. Let $\vec{b} = 4i + 3j$ and $\vec{c}$ be two vectors perpendicular to each other in the xy – plane. If $\vec{r}_i$, $i = 1, 2, ...n$ are the vectors in the same plane having projection 1 and 2 along b and c respectively $\sum_{i=1}^{n} \left|\vec{r}_i\right|^2$ is equal to k then $\frac{k + 20}{16}$ equals to

51. The smallest value of 'a' for which the inequality $(a-1)x^2 - (a+1)x + (a-1) \geq 0$ holds for all $x \geq 2$ is k then the value of $\frac{3k}{14}$ is

52. Three distinct dice are thrown and the sum of the numbers appearing on the top faces is 12. If n is the number of ways in which this is possible, the sum of the digits of n is equal to

53. If roots of $x^3 + ax^2 + bx - \frac{1}{9} = 0$ are in A.P. the $\frac{121}{16}(2a^3 - 9ab + 1)$ is _____.

54. Let A and B be two non singular matrices of order 2 such that $9A^2B - 6AB + B = 9\left\{B - \begin{bmatrix} 2 & -3 \\ 2 & 1 \end{bmatrix}\right\}$. If $B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}$ and Trace (A) $(A) = \frac{10}{3}$ then $\frac{|A|}{4}$ is